AI-Driven Flood Susceptibility Mapping for Arid Regions: A UAE Case Study

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1. Introduction

Flooding in arid regions, including the UAE, poses increasing risks due to climate change, urbanization, and extreme rainfall events. Traditional flood susceptibility mapping relies on hydrological models requiring extensive local data. This study presents an **AI-driven approach** using terrain features derived from remote sensing data to predict flood susceptibility in the United Arab Emirates.

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2. Methodology

Data Sources:

- Digital Elevation Model (SRTM/Copernicus) – 30m resolution
- River network (OpenStreetMap)

Feature Engineering (6 Terrain Features):

1. **Elevation** – Primary determinant of water flow direction
2. **Slope** – Steeper slopes reduce infiltration, increase runoff
3. **Aspect** – Orientation affecting moisture retention
4. **Curvature** – Convergence/divergence of water flow
5. **Distance_to_River** – Proximity to drainage networks (most important)
6. **Topographic Wetness Index (TWI)** – Potential for water accumulation

Machine Learning:

- **Algorithm**: Random Forest Classifier
- **Dataset**: 10,000 samples (7,000 training, 3,000 testing)
- **Validation**: 5-fold cross-validation

Deployment:

- Flask web application (CSV upload → visualization → PDF report)
- Coordinates-based system (KD-tree indexing, 70% complete)

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3. Results

Metric	Value
Training Accuracy	99.4%
Testing Accuracy	99.5%
AUC Score	1.000
Cross-Validation (Mean)	0.994 (±0.001)
Flood Class Precision	1.00
Flood Class Recall	0.99
Flood Class F1	1.00

Feature Importance (Top 3):

1. **Distance_to_River** (45%) – Most critical factor
2. **Elevation** (30%)
3. **Slope** (12%)

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4. Key Findings

- **UAE's arid terrain** shows high flood susceptibility near wadi systems and urban low-lying areas.
- **Distance to river** is the single most important predictor of flood risk.
- **Machine learning** can effectively identify flood-prone areas without complex hydrological models.
- **Web-based deployment** enables rapid risk assessment for urban planning, real estate, and disaster preparedness.

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5. Applications & Impact

Application	Benefits
Urban Planning	Identify flood-prone areas for development restrictions
Real Estate	Property risk assessment (AED 1,800/report)
Disaster Preparedness	Early warning systems and evacuation planning
Climate Adaptation	Support UAE's COP28 climate goals

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6. Next Steps

- ✅ **Completed**: Model training, Flask app, PDF reporting
- 🔄 **In Progress**: Coordinates-based input system (70% complete)
- 📋 **Planned**: Integration with real-time rainfall data
- 📋 **Planned**: Cloud deployment (AWS/Azure)

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7. Alignment with UAEU Research Priorities

This project aligns with UAEU's focus on:

- **Climate adaptation and disaster risk management**
- **Remote sensing and GIS applications**
- **AI/ML for environmental sustainability**
- **UAE-specific research addressing regional challenges**

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8. Deliverables

Deliverable	Status
Trained Random Forest Model	✅ Complete
Flask Web Application	✅ Complete
Automated PDF Reports	✅ Complete
Video Demonstration (3 min)	✅ Complete
GitHub Repository	✅ Complete
Abstract	✅ Complete
Coordinates-Based System	🔄 In Progress

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9. Relevant Skills Demonstrated

Skill Area	Tools & Methods
Remote Sensing	DEM, SRTM, Copernicus, rasterio, GDAL, QGIS
Machine Learning	Random Forest, scikit-learn, cross-validation, AUC
Python Programming	pandas, numpy, matplotlib, Flask, reportlab
Web Development	Flask, HTML, CSS, PDF generation
Geospatial Analysis	TWI, distance-to-river, coordinate transformations

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Keywords: Flood Susceptibility, AI, Remote Sensing, Machine Learning, Disaster Risk Management, UAE, Climate Adaptation